

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin 6

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RSA-Encryption

Let Alice have the primes $p = 107$, $q = 127$ and the exponent $k = 5$.

Determine m and $\Phi(m)$ and show that $k = 5$ is a valid choice by computing $\gcd(k, \Phi(m))$.

from chapter 16.1 [1] it follows that $m = p \cdot q = 107 \cdot 127 = 13589$. Then, $\Phi(m) = (p - 1)(q - 1) = 106 \cdot 126 = 13356$.

Computing $\gcd(k, \Phi(m)) = \gcd(5, 13356)$, since k and $\Phi(m)$ needs to be coprime for k to be a valid choice.

This can be computed using the Euclidean algorithm 3.2 [1]: let $r_0 = 13356$, $r_1 = 5$, and compute the sequence of remainders:

$$\begin{aligned} 13356 &= q_1 \cdot 5 + 1 \quad \text{for } q_1 = 2671 \text{ and } r_2 = 1 \\ 5 &= q_2 \cdot 1 + 0 \quad \text{for } q_2 = 5 \text{ and } r_3 = 0 \end{aligned}$$

hence

$$\gcd(5, 13356) = r_2 = 1$$

since the greatest common divisor is 1, according to definition 3.1 [1] then $k = 5$ is coprime to $\Phi(m) = 13356$ and is a certified valid choice for a public key.

Determine Alice's private exponent u

A number u is a valid private exponent if it satisfies the equation $k \cdot u \equiv 1 \pmod{\Phi(m)}$.

To determine this, see definition 9.18 [1] to see how to determine such a number u .

Some prerequisites to use the method in definition 9.18 [1] is to know the value of $\Phi(m)$ for $m = p \cdot q$, $p \neq q$ where p and q are primes, and that k is coprime to $\Phi(m)$. All of these prerequisites are satisfied in the previous section.

The method uses the extended Euclidean algorithm 3.3 [1] to find bezout coefficients u and v such that $k \cdot u + \Phi(m) \cdot v = 1$.

Use the algorithm to find u :

let $ku + \Phi(m)v = \gcd(k, \Phi(m))$, where $k = 5$ and $\Phi(m) = 13356$. the quotients $q_1 = 2671$ and $q_2 = 5$ are already computed in the previous section.

i	r_i	q_i	u_i	v_i
0	13356	—	0	1
1	5	2671	1	0
2	1	5	-2671	1

Table 1: Extended Euclidean algorithm for 5 and $\Phi(m) = 13356$

sanity check:

$$\begin{aligned} k \cdot u + \Phi(m) \cdot v &= 1 \\ 5 \cdot (-2671) + 13356 \cdot 1 &= 1 \end{aligned}$$

even though the current $u = -2671$ is a valid solution, in accordance to definition 9.18 [1] it needs $u > 0$ and $v < 0$.

To accomodate this, let $v' = v \cdot -1 = -1$ and $u' = \Phi(m) - u = 10685 = 13356 - 2671$, and then double check that it's still valid:

$$\begin{aligned} k \cdot u' + \Phi(m) \cdot v' &= 1 \\ 5 \cdot 10685 + 13356 \cdot (-1) &= 1 \end{aligned}$$

which it still is, hence $u' = 10685$ and $v' = -1$ is a valid solution.

Then it follow from equation 9.13 [1]:

$$ku' + \Phi(m)v' = 1 \implies ku' = 1 - \Phi(m)v' \implies ku' = 1 + \Phi(m)w$$

for $w = -v'$ and letting the multiplier be a modulo $\Phi(m)$, then it follows that

$$ku' = 1 + \text{mod } \Phi(m)w \implies ku' \equiv 1 \text{ mod } \Phi(m)$$

and for $u' = 10685$ and the rest of the values

$$5 \cdot 10685 \equiv 1 \text{ mod } 13356$$

and since u' was found via the extended Euclidean algorithm, then $u' = 10685$ is a valid solution to the equation $ku' \equiv 1 \text{ mod } \Phi(m)$, which means that $u' = 10685$ is a valid private exponent.

References

- [1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.